

ANTHROPIC INFLUENCE ON THE DECLINE OF NATURAL SPACES: ANALYSIS OF LAND COVER AND LAND USE CHANGE IN THE WARAIRA REPANO NATIONAL PARK, 1985-2023

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RESUMEN

This study addressed the anthropic pressure on the Waraira Repano National Park (PNWR), quantifying the impact of land use change on its ecosystems in the 1985–2023 period. The main objective of the research was to determine the annual replacement rate of natural spaces by human uses, identifying the direct correlation with the expansion of agricultural use. The methodology employed the annual historical series from MapBiomass Venezuela, reclassifying the land covers into two symmetrical macro-classes (Natural Spaces and Anthropic Uses). The Python programming language was used to apply a linear regression model and Pearson correlation analysis to the 39-year time series. The results revealed a net land cover transfer of 1.4%, equivalent to the loss of 1,180 hectares of natural spaces. The statistical model confirmed a perfect negative correlation ($r=-1.000$), establishing that the expansion operates under a direct replacement principle, with a constant rate of 48.1 hectares per year (ha/year). Agricultural Use was identified as the main driver of this pressure. It is concluded that this inertial trend ($R^2=0.821$), which projects an accumulated invasion of 4,660 ha by the year 2040, is a sustained threat that compromises the capacity of Waraira Repano as an essential water source for the region, for which conservation management by government entities based on precise geospatial data is proposed.

1. Introduction

1.1 Objectives

The Waraira Repano National Park, commonly known as El Ávila, is the main green lung of Caracas. Its ecological function is fundamental for climate stability, biodiversity conservation, and the provision of water resources that supply the Venezuelan capital. Despite having the status of a protected area, the park faces constant pressure from the unregulated expansion of anthropic activities in its buffer zones and slopes.

In this paper, we aim to demonstrate the negative impact that land use change represents on the park's ecosystems, focusing particularly on the expansion of agricultural use.

The main objective of this research **is to quantify the annual replacement rate of natural spaces by anthropic uses in the Waraira Repano NP (1985-2023)**, identifying the direct correlation between the loss of natural formations and the increase in agricultural uses during this period.

1.2 Historical/Social Context

The **Waraira Repano National Park** was officially declared in 1958. It is located within the Cordillera de la Costa (Coastal Mountain Range), extending across Caracas, the north of Miranda state, and the south of La Guaira state. It has a surface area of 85,192 hectares (ha).

The most important function of the Waraira Repano is **hydrological regulation**, acting as a natural water source. Its dense forest ecosystem is essential for water capture and filtration, serving as the birthplace of numerous streams and ravines fundamental for the regional water supply. These include systems such as La Quebrada de Cotiza, Chacaito, Anauco, Las Adjuntas, El Sebucán, Los Chorros (El Paraíso), and La Julia, among many others.

This water supply network is vital for regional sustainability. The waterways on the Northern slope provide their flows to the towns of the Central Coast, including La Guaira, Naiguatá, and Los Caracas. Meanwhile, on the Southern slope, the park supplies water directly to the metropolis of Caracas, as well as intermediate cities like Guatire and Guarenas. Additionally, rivers such as Río Grande and Río Chiquito have been a source of electrical energy for the region, reaffirming the strategic value and critical importance

2. Methodology

For this analysis, the annual historical land cover and land use series provided by **MapBiomass Venezuela** (1985–2023) are used. The robustness of these geospatial data, covering a 39-year period, allows moving from qualitative observation to precise quantification of territorial transformations.

To validate the human pressure hypothesis, MapBiomass' detailed categories were reclassified into two main macro-classes to simplify the replacement flow analysis:

Macro-Class	MapBiomass Categories
Natural Spaces	Forest Formations, Savanna/Grassland, Shrubland, River, lake or ocean
Anthropic Uses	Agricultural Use, Urban Use, Mining Use, Areas without Vegetation

This way of grouping the data is fundamental to measure the net loss of natural capital against the expansion of human intervention in the park's territory throughout the time series.

The data analysis was entirely developed using **the Python programming language** and the following libraries: pandas for time series manipulation and structuring, matplotlib for visualization, and scikit-learn for modeling.

A) Linear Regression Model (Quantification of the Replacement Rate)

A multivariate linear regression model was applied to quantify the specific influence of human activities on the decline of natural cover. The methodology focused on:

1. Dependent Variable (y): Annual surface area of Natural Spaces.
2. Independent Variables (X): Annual surface areas of Agricultural Use, Urban Use, and Mining Use.

The model allows obtaining the regression coefficients (β), which indicate the contribution (natural hectares lost per hectare gained in a specific use) of each human use. Of particular interest is the coefficient associated with Agricultural Use, which will serve to validate the correlation proposed in the objective. Furthermore, the coefficient of determination (R^2) will indicate the percentage of the total variance in natural loss that is explained by anthropic expansion.

B) Time Series Analysis

The time series from 1985 to 2023 was used to generate a stacked area chart visualizing the historical trend and the magnitude of change in the balance between the Natural Spaces and Anthropic Uses macro-classes.

To complement the quantitative evidence, MapBiomass' cartographic resources were used to generate two contrasting maps that spatially illustrate the transformation: the Initial Map (1985) and the Final Map (2023). The visual comparison allows identifying geographically the areas of greatest advancement of the agricultural and urban frontiers within the park's perimeter.

3. Results

3.1. Historical Balance and Change Quantification

The analysis of the 1985-2023 period over the total area of the Waraira Repano NP (81,763 ha) reveals a net land cover transfer from natural spaces to anthropic uses.

Category	Initial Area (1985)	Final Area (2023)	Net Change (ha)	Net Change (%)
Natural Spaces	79.380 ha (97.1%)	78.200 ha (95.6%)	-1.180 ha	-1.4%
Human Uses	2.383 ha (2.9%)	3.562 ha (4.4%)	+1.180 ha	+1.4%

It is confirmed that the total area of the park remained constant (standard deviation of 0.35 ha), indicating that the loss of 1,180 hectares of natural spaces correlates exactly with the gain of 1,180 hectares of human uses.

3.2. Regression Analysis and Replacement Rate

Applying the Linear Regression model to the 38-year time series yielded the following key parameters:

- **Annual Rate of Change:** The model predicts that Natural Spaces decreased at a constant rate of -48.1 ha/year, while Human Uses increased at a symmetrical rate of +48.1 ha/year.
- **Coefficient of Determination (R^2):** An $R^2=0.821$ was obtained for both trends. This high value indicates that **82.1% of the variance** in area loss/gain is simply explained

by the linear progression of time, confirming a very stable and predictable trend of change throughout the period..

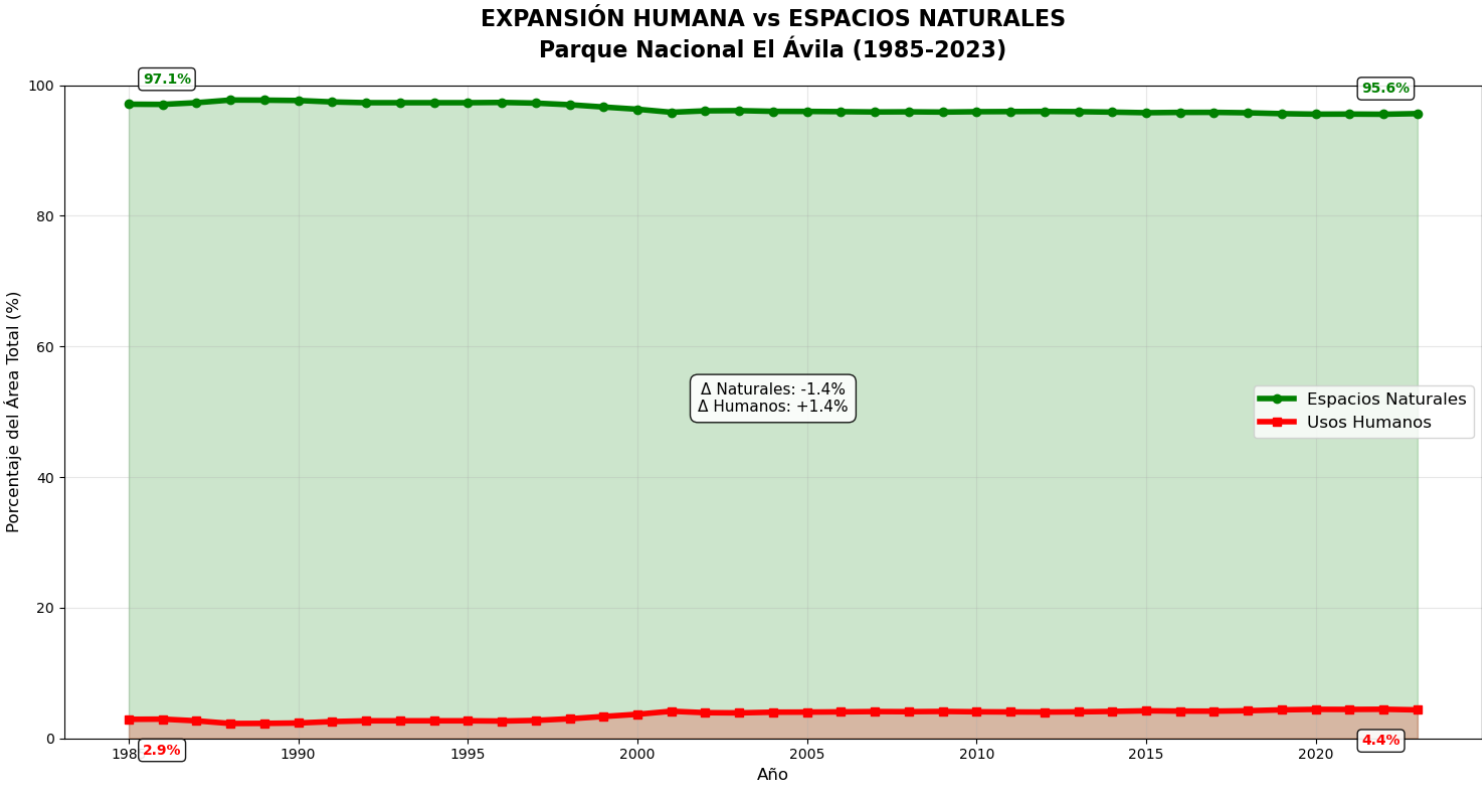
3.3. Statistical Correlation (Definitive Validation)

The Pearson correlation analysis between the Natural Spaces and Human Uses series yielded a value of $r=-1.000$ with a p-value of 0.000000.

This statistical result represents a **perfect negative correlation**, constituting **conclusive evidence** that the expansion of human activities in the Waraira Repano NP operates under a **direct replacement** principle: every hectare gained by an anthropic use is a lost hectare of natural cover.

3.3. Analysis of computer-generated graphs in Python

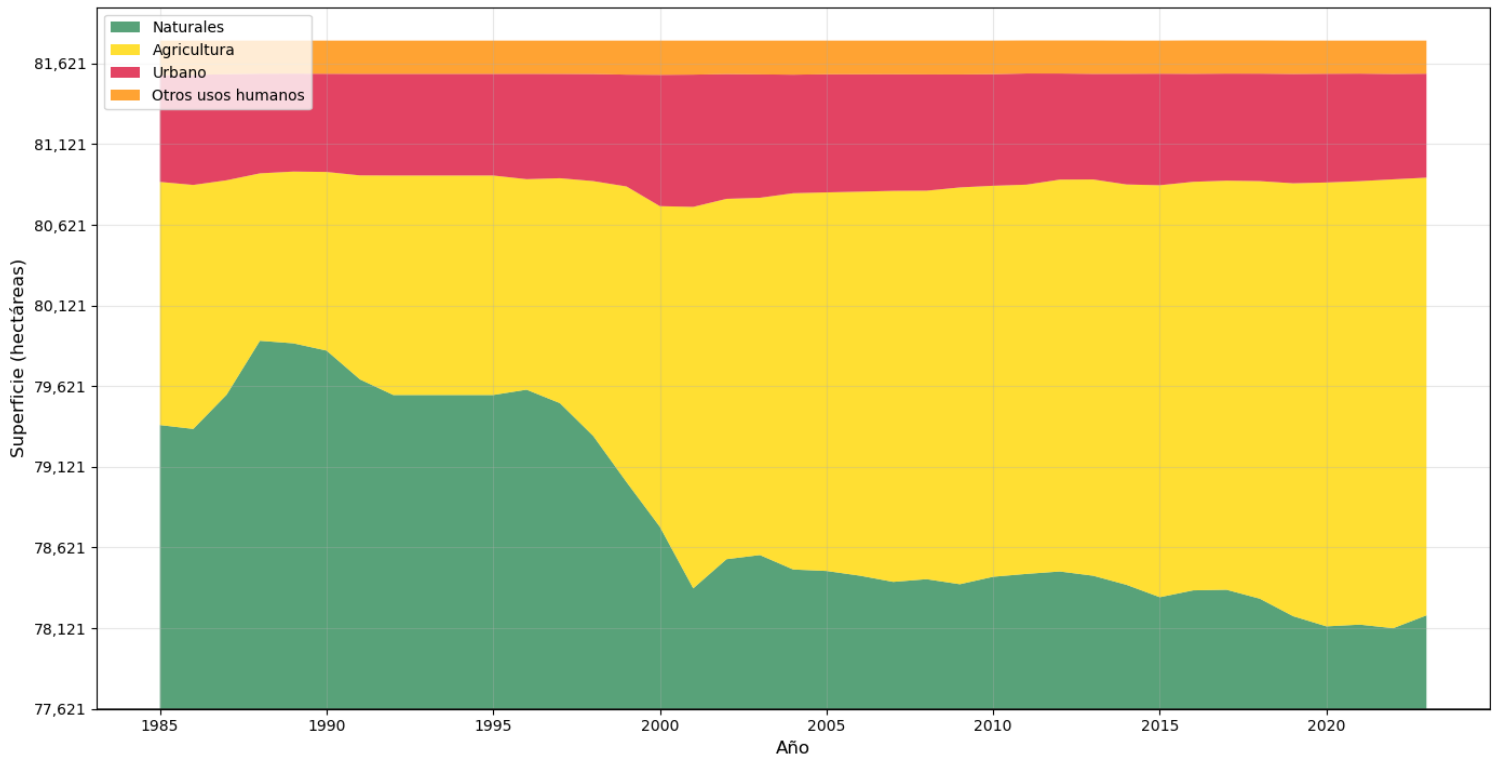
Graph 1: Evolution of percentages with shaded lines.



In this graph, we can observe that as the green line (Natural Spaces) steadily descends, the red line (Human Uses) ascends. The gain/loss separation rate is 1.4%, clear evidence of land cover transfer and symmetry.

Graph 2: Stacked areas.

EXPANSIÓN HUMANA - COMPOSICIÓN DE USOS Parque Nacional El Ávila (1985-2023)



In this graph, we have the following layers:

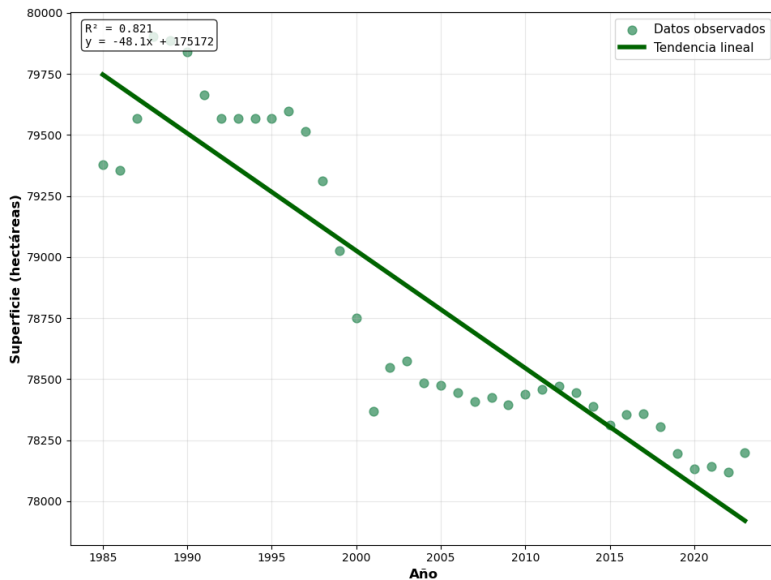
- Green: Natural spaces, which include forest formations, savanna/grassland, shrubland, river, lake, or ocean.
- Yellow: Agricultural use.
- Red: Urban use, such as buildings and housing.
- Orange: Other human uses, for example, mining use.

The data are shown from an adjusted baseline (**77,620.77 ha**) to better perceive the changes. The visualization confirms that Agricultural Use is the layer with the largest extension and upward growth, acting as the main driver of the constant pressure. Urban Use and Other Human Uses remain almost constant, which directs the analysis' attention to the agricultural frontier.

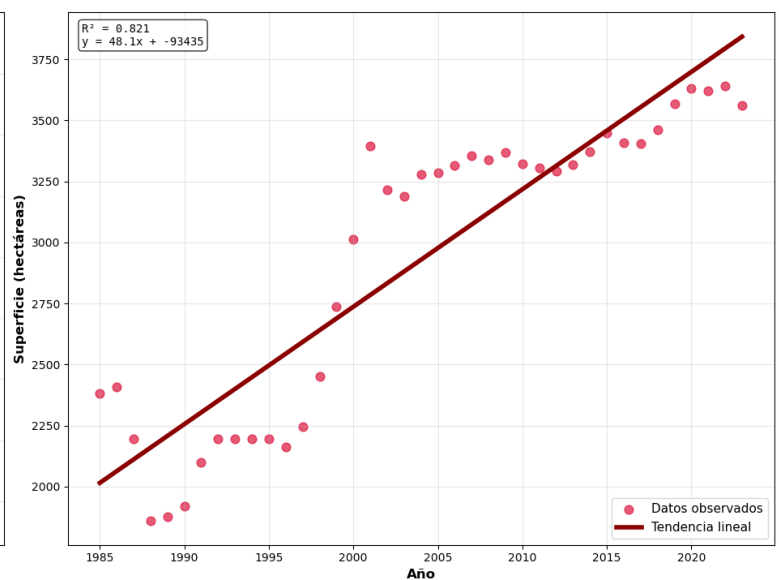
”

Graph 3: Linear Regressions.

REGRESIÓN LINEAL: Espacios Naturales
(1985-2023)



REGRESIÓN LINEAL: Usos Humanos
(1985-2023)



In this scatter plot, the trend representing the pattern of the observed data can be visualized. In this case, each year of the historical line 1985-2023 is represented as a point on each graph. The green graph represents the linear regression of natural spaces and the red graph the linear regression of human uses (agricultural use, urban use, and any other type of human use).

The coefficient of determination, which measures the goodness of fit of the regression models, results in 82.1% ($R^2 = 0.821$), confirming a very stable and predictable trend of change over the period.

Obtenemos las ecuaciones de la recta de la tendencia lineal

- **Equation for Natural Spaces:** $y = -48.1x + 95484$
 - Slope ($\beta_1 = -48.1$): It means that Natural Spaces decrease on average by 48.1 hectares for every passing year.
- **Equation for Human Uses:** $y = +48.1x - 13721$
 - Slope ($\beta_1 = +48.1$): This symmetrical and positive slope confirms direct replacement. It means that Human Uses increase on average by 48.1 hectares for every passing year.

A perfect symmetry between both graphs is observed, obtaining a clear replacement relationship of natural spaces by human use spaces.

Maintaining the current linear trend of -48.1 ha/year, the projected loss of natural cover is as follows:

- The rate of loss, although moderate in percentage terms, **is constant, predictable, and statistically significant**, implying a sustained threat to the ecological integrity of the Waraira Repano NP.

Esta sección esta destinada a la comparación visual del PN Warairarepano 1985 vs 2023. Con extraídos de la herramienta Biomap Venezuela.

The replacement of natural spaces by agricultural use is observable by noticing the beige patches. The increase in agricultural use is especially found in the northwestern part of the national park, in La Guaira state.

4. Discussion and Conclusions

The results of this study validate the initial hypothesis, demonstrating the land use transition within the Waraira Repano NP. The linear regression and Pearson correlation coefficient of $r = -1.000$ constitute conclusive evidence that the expansion of human uses in the park operates under a **direct replacement** principle over the natural zones of forest formations, savanna/grassland, shrubland, rivers, or lakes.

Although the net loss of 1.4% seems modest in percentage terms, the constant rate of 48.1 ha of annual invasion is a sustained threat that compromises the park's integrity in the long term. The high stability of the trend, demonstrated by the $R^2=0.821$, underscores that this pressure is an inertial and structural process.

The observed trend of change, driven mainly by agricultural expansion (as evidenced by the Stacked Area Chart), has critical implications that go beyond the surface area:

1. **Ecological and Water Risk:** The constant loss of 48.1 ha of vegetation cover annually reduces the park's capacity to capture and regulate water, directly affecting the ecosystem services that benefit the population of Caracas, and increasing vulnerability to extreme weather events.
2. **Inertia of Degradation:** The projection to the year 2040, which estimates an invasion of 4,660 ha of Human Uses, indicates that without intervention, the degradation process will continue predictably, reaching a point of no return.

The quantitative analysis from MapBiomass and the regression model provide a technical and numerical basis for park management. The solution lies not only in stopping the loss but in intelligently managing the trend. Given that agricultural activity is the main driver of expansion, **government control** and awareness policies must be more rigorous in agricultural frontier zones, offering sustainable economic alternatives to communities that do not involve degrading the protected area. It is urgent that conservation decisions migrate from intention to **data-driven action**, using tools like **MapBiomass** to ensure the ecological integrity and water future of the region.

5. Used Code:

In the following section is the Python code used.

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3 from sklearn.linear_model import LinearRegression
4 import numpy as np
5 from scipy import stats
6
7 # --- 1. CARGAR Y PREPARAR DATOS ---
8 try:
9     df = pd.read_csv(filepath_or_buffer='datos_biomap_csv.csv', encoding='latin-1')
10 except FileNotFoundError:
11     print("Error: El archivo 'datos_biomap_csv.csv' no se encontró.")
12     exit()
13
14 # --- DEFINIR CATEGORÍAS ---
15 # Espacios Naturales
16 df['Naturales'] = df['Bosque'] + df['Sabana/Herbazal'] + df['Arbustal'] + df['Río, lago u oceano']
17 # Usos Humanos
18 df['Antropogenicos'] = df['Uso agropecuario'] + df['Uso urbano'] + df['Uso minero'] + df['Sin vegetacion']
19
20 # Verificación de datos
21 df['Total_verificado'] = df['Naturales'] + df['Antropogenicos']
22 variacion_total = df['Total_verificado'].std()
23 print(f"✅ Verificación: Desviación del área total = {variacion_total:.2f} ha")
24
25 # Calcular porcentajes
26 df['_Naturales'] = (df['Naturales'] / df['Total_verificado']) * 100
27 df['_Antropogenicos'] = (df['Antropogenicos'] / df['Total_verificado']) * 100
28
29 # --- 2. GRÁFICO 1: EVOLUCIÓN DE PORCENTAJES (Lineas con sombreado) ---
30 plt.figure(figsize=(15, 8))
31
32 # Gráfico de líneas principales
33 plt.plot(*args: df['Year'], df['_Naturales'], color='green', linewidth=4, label='Espacios Naturales', marker='o', markersize=6)
34 plt.plot(*args: df['Year'], df['_Antropogenicos'], color='red', linewidth=4, label='Usos Humanos', marker='s', markersize=6)
35
36 # Área sombreada
37 plt.fill_between(df['Year'], df['_Naturales'], alpha=0.2, color='green')
38 plt.fill_between(df['Year'], df['_Antropogenicos'], alpha=0.2, color='red')
39
40 # Personalización
41 plt.title(label='EXPANSIÓN HUMANA vs ESPACIOS NATURALES\nParque Nacional El Ávila (1985-2023)',
```

```

43 plt.ylabel(ylabel: 'Porcentaje del Área Total (%)', fontsize=12)
44 plt.xlabel(xlabel: 'Año', fontsize=12)
45 plt.legend(fontsize=12, loc='best')
46 plt.grid(visible: True, alpha=0.3)
47 plt.ylim(*args: 0, 100)
48
49 # Añadir valores
50 valores = [
51     (df['Year'].iloc[0], df['_Naturales'].iloc[0], 'green', (10, 15), f"{df['_Naturales'].iloc[0]:.1f}%"),
52     (df['Year'].iloc[-1], df['_Naturales'].iloc[-1], 'green', (-40, 15), f"{df['_Naturales'].iloc[-1]:.1f}%"),
53     (df['Year'].iloc[0], df['_Antropogenicos'].iloc[0], 'red', (10, -25), f"{df['_Antropogenicos'].iloc[0]:.1f}%"),
54     (df['Year'].iloc[-1], df['_Antropogenicos'].iloc[-1], 'red', (-40, -25), f"{df['_Antropogenicos'].iloc[-1]:.1f}%")
55 ]
56
57 for x, y, color, offset, texto in valores:
58     plt.annotate(texto, xy=(x, y), xytext=offset, textcoords='offset points',
59                 color=color, fontweight='bold', fontsize=10,
60                 bbox=dict(boxstyle="round,pad=0.3", facecolor="white", alpha=0.9))
61
62 # Añadir cambio neto
63 cambio_nat = df['_Naturales'].iloc[-1] - df['_Naturales'].iloc[0]
64 cambio_hum = df['_Antropogenicos'].iloc[-1] - df['_Antropogenicos'].iloc[0]
65
66 plt.text(df['Year'].mean(), y: 50,
67         s: f'Δ Naturales: {cambio_nat:+.1f}%\nΔ Humanos: {cambio_hum:+.1f}%',
68         bbox=dict(boxstyle="round,pad=0.5", facecolor="white", alpha=0.9),
69         ha='center', fontsize=11)
70
71 plt.tight_layout()
72 plt.show()
73
74 ##END
75
76 # --- 3. GRÁFICO 2: ÁREA APILADA ---
77 min_naturales = df['_Naturales'].min() # 500 ha desde el base_line
78 base_line_optima = min_naturales - 500 # A partir se empieza a mostrar la grafica 77,620.77 ha
79
80 plt.figure(figsize=(14, 8))

```

```

80 plt.figure(figsize=(14, 8))
81
82 # Ajustar los datos
83 naturales_ajustado = df['Naturales'] - base_line_optima
84 agropecuario_ajustado = df['Uso agropecuario']
85 urbano_ajustado = df['Uso urbano']
86 otros_ajustado = df['Uso minero'] + df['Sin vegetacion']
87
88 plt.stackplot(df['Year'],
89              *args: [naturales_ajustado, agropecuario_ajustado, urbano_ajustado, otros_ajustado],
90              labels=['Naturales', 'Agricultura', 'Urbano', 'Otros usos humanos'],
91              colors=['#2E8B57', '#FFD700', '#DC143C', '#FF8C00'],
92              alpha=0.8)
93
94 plt.axhline(y=0, color='black', linestyle='-', alpha=0.7, linewidth=2)
95 plt.title(label: f'EXPANSIÓN HUMANA - COMPOSICIÓN DE USOS\nParque Nacional El Ávila (1985-2023)',
96          fontsize=16, fontweight='bold', pad=20)
97 plt.ylabel(ylabel: f'Superficie (hectáreas)', fontsize=12)
98 plt.xlabel(xlabel: 'Año', fontsize=12)
99 plt.legend(loc='upper left')
100 plt.grid(visible: True, alpha=0.3)
101
102 # Formatear eje Y
103 plt.gca().yaxis.set_major_formatter(plt.FuncFormatter(lambda x, p: f'{x + base_line_optima:,.0f}'))
104 plt.tight_layout()
105 plt.show()
106
107 ##END
108
109 # --- 4. REGRESIÓN LINEAL (preparacion de datos)---
110 # Preparar datos para regresión (evitar warnings)
111 X = df['Year'].values.reshape(-1, 1)
112 y_naturales = df['Naturales'].values
113 y_humanos = df['Antropogenicos'].values
114
115 # Modelos de regresión
116 modelo_naturales = LinearRegression()
117 modelo_humanos = LinearRegression()
118

```

```

119 modelo_naturales.fit(X, y_naturales)
120 modelo_humanos.fit(X, y_humanos)
121
122 prediccion_naturales = modelo_naturales.predict(X)
123 prediccion_humanos = modelo_humanos.predict(X)
124
125 # --- . GRÁFICO 3: GRÁFICOS DE REGRESIÓN LINEAL (Subplots) ---
126
127 # Gráficos de regresión
128 fig, (ax1, ax2) = plt.subplots( nrows= 1, ncols= 2, figsize=(20, 8))
129
130 # Sub Gráfico 1: Espacios Naturales
131 ax1.scatter(df['Year'], y_naturales, label='Datos observados', color='#2E8B57', alpha=0.7, s=60)
132 ax1.plot(df['Year'], prediccion_naturales, color='darkgreen', linewidth=4, label='Tendencia lineal')
133 ax1.set_title('REGRESIÓN LINEAL: Espacios Naturales\n(1985-2023)', fontsize=16, fontweight='bold', pad=15)
134 ax1.set_xlabel('Año', fontsize=12, fontweight='bold')
135 ax1.set_ylabel('Superficie (hectáreas)', fontsize=12, fontweight='bold')
136 ax1.legend(fontsize=11)
137 ax1.grid(True, alpha=0.3)
138
139 r2_naturales = modelo_naturales.score(X, y_naturales)
140 ecuacion_nat = f'y = {modelo_naturales.coef_[0]:.1f}x + {modelo_naturales.intercept_[0]:.0f}'
141 ax1.text(0.02, 0.98, f'R² = {r2_naturales:.3f}\n{ecuacion_nat}',
142         transform=ax1.transAxes, verticalalignment='top',
143         bbox=dict(boxstyle='round', facecolor='white', alpha=0.8),
144         fontsize=10, fontfamily='monospace')
145
146 # Sub Gráfico 2: Usos Humanos
147 ax2.scatter(df['Year'], y_humanos, label='Datos observados', color='#DC143C', alpha=0.7, s=60)
148 ax2.plot(df['Year'], prediccion_humanos, color='darkred', linewidth=4, label='Tendencia lineal')
149 ax2.set_title('REGRESIÓN LINEAL: Usos Humanos\n(1985-2023)', fontsize=16, fontweight='bold', pad=15)
150 ax2.set_xlabel('Año', fontsize=12, fontweight='bold')
151 ax2.set_ylabel('Superficie (hectáreas)', fontsize=12, fontweight='bold')
152 ax2.legend(fontsize=11)
153 ax2.grid(True, alpha=0.3)

```

```

154 r2_humanos = modelo_humanos.score(X, y_humanos)
155 ecuacion_hum = f'y = {modelo_humanos.coef_[0]:.1f}x + {modelo_humanos.intercept_:0f}'
156 ax2.text(0.02, 0.98, f'R² = {r2_humanos:.3f}\n{ecuacion_hum}',
157         transform=ax2.transAxes, verticalalignment='top',
158         bbox=dict(boxstyle='round', facecolor='white', alpha=0.8),
159         fontsize=10, fontfamily='monospace')
160
161 plt.tight_layout()
162 plt.show()
163
164 # --- 5. ANÁLISIS COMPLETO ---
165 print("\n" + "="*70)
166 print("📊 ANÁLISIS COMPLETO: EXPANSIÓN HUMANA vs ESPACIOS NATURALES")
167 print("="*70)
168
169 # Cálculos básicos
170 cambio_absoluto_nat = df['Naturales'].iloc[-1] - df['Naturales'].iloc[0]
171 cambio_absoluto_ant = df['Antropogenicos'].iloc[-1] - df['Antropogenicos'].iloc[0]
172 años_totales = df['Year'].iloc[-1] - df['Year'].iloc[0]
173
174 print(f"\n • PERIODO ANALIZADO: {df['Year'].iloc[0]} - {df['Year'].iloc[-1]} ({años_totales} años)")
175 print(f" • ÁREA TOTAL: {df['Total_verificado'].mean():0f} ± {variacion_total:.1f} ha")
176
177 print(f"\n 🌱 ESPACIOS NATURALES:")
178 print(f" • 1985: {df['Naturales'].iloc[0]:0f} ha ({df['%_Naturales'].iloc[0]:.1f}%)")
179 print(f" • 2023: {df['Naturales'].iloc[-1]:0f} ha ({df['%_Naturales'].iloc[-1]:.1f}%)")
180 print(f" • Cambio: {cambio_absoluto_nat:+0f} ha (Δ{cambio_nat:+.1f}%)")
181 print(f" • Tendencia: {modelo_naturales.coef_[0]:.1f} ha/año (R² = {r2_naturales:.3f})")
182
183 print(f"\n 🏡 USOS HUMANOS:")
184 print(f" • 1985: {df['Antropogenicos'].iloc[0]:0f} ha ({df['%_Antropogenicos'].iloc[0]:.1f}%)")
185 print(f" • 2023: {df['Antropogenicos'].iloc[-1]:0f} ha ({df['%_Antropogenicos'].iloc[-1]:.1f}%)")
186 print(f" • Cambio: {cambio_absoluto_ant:+0f} ha (Δ{cambio_hum:+.1f}%)")
187 print(f" • Tendencia: {modelo_humanos.coef_[0]:.1f} ha/año (R² = {r2_humanos:.3f})")
188
189 # Correlación y significancia
190 correlacion, p_valor = stats.pearsonr(df['Naturales'], df['Antropogenicos'])
191

```



```

192 print(f"\n🔍 RELACIÓN ESTADÍSTICA:")
193 print(f"    • Correlación: {correlacion:.3f}")
194 print(f"    • Valor-p: {p_valor:.6f}")
195
196
197 # Interpretación
198 if correlacion < -0.8 and p_valor < 0.05:
199     print("    ✅ FUERTE EVIDENCIA: Expansión humana REEMPLAZA directamente espacios naturales")
200 elif correlacion < -0.6 and p_valor < 0.05:
201     print("    ✅ EVIDENCIA MODERADA: Hay reemplazo significativo")
202 else:
203     print("    ⚠️ EVIDENCIA MODERADA-BAJA: Revisar otros factores")
204
205 # Proyecciones futuras (sin warnings)
206 print(f"\n📅 PROYECCIÓN (tendencia actual):")
207 future_years = np.array([2025, 2030, 2040]).reshape(-1, 1)
208 proy_nat = modelo_naturales.predict(future_years)
209 proy_hum = modelo_humanos.predict(future_years)
210
211 for i, year in enumerate([2025, 2030, 2040]):
212     print(f"    • {year}: Naturales = {proy_nat[i]:.0f} ha, Humanos = {proy_hum[i]:.0f} ha")
213
214 # Resumen ejecutivo
215 print("\n" + "="*70)
216 print("📌 RESUMEN EJECUTIVO")
217 print("="*70)
218
219 if cambio_nat < -2: # Si pérdida mayor al 2%
220     print("    ▼ CONCLUSIÓN: EXPANSIÓN HUMANA SIGNIFICATIVA")
221     print(f"    • {abs(cambio_absoluto_nat):.0f} ha de naturales perdidas")
222     print(f"    • {cambio_absoluto_ant:.0f} ha de usos humanos ganadas")
223     print(f"    • {abs(cambio_nat):.1f}% del territorio transformado")
224     print(f"    • Tasa: {abs(modelo_naturales.coef_[0]):.1f} ha/año de pérdida natural")
225 else:
226     print("    ♦ CONCLUSIÓN: CAMBIO MODERADO")
227
228 print("="*70)

```


6. Bibliography

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